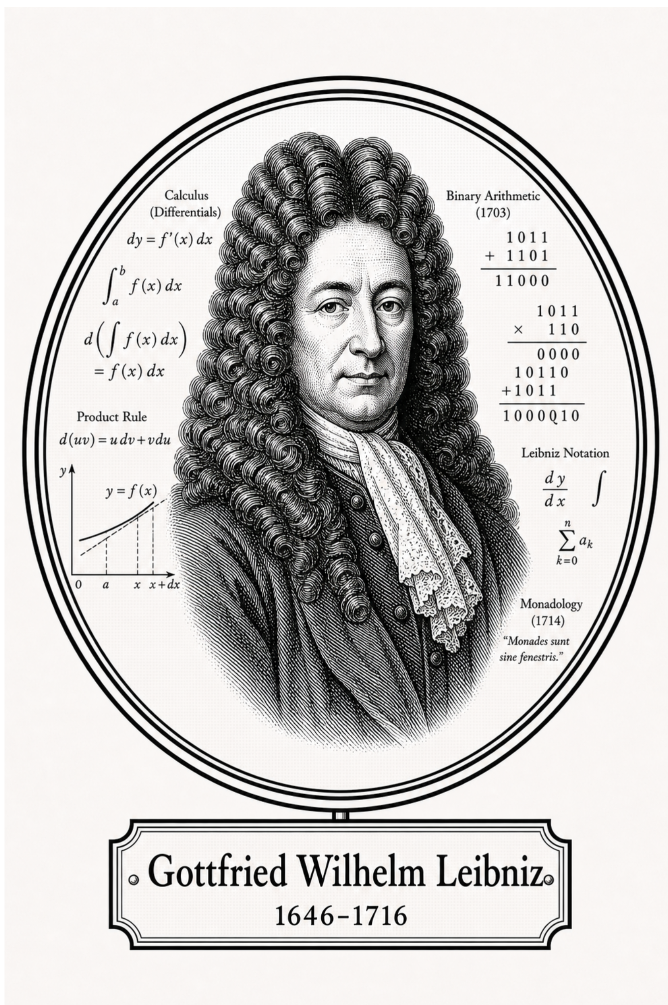
From Cantor to Ito

Classical Analysis

Differentiation



Gottfried Wilhelm Leibniz

1646-1716

Self-Study Notes

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For every reader learning to make mathematics their own.

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1.1 Derivative Definition

Derivative Definition — toolkit

Concept family: The derivative at a point, and its three equivalent formulations.

Atomic items in this section:

- Definition: Derivative at a Point (epsilon–delta form)
- Definition: Topological Definition of Derivative at a Point
- Definition: Sequential Definition of Derivative at a Point
- Theorem: Equivalence of Definitions of the Derivative at a Point
- Proposition: Derivative: h-Form Equivalence
- Theorem: Differentiable Implies Continuous
- Theorem: Uniqueness of the Derivative
- Proposition: Derivative of the Identity

Definition (Derivative at a Point)

Definition 1.1.1 (Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ be a real-valued function and $c \in \operatorname{int}(D)$. We say f is *differentiable at* c with *derivative* $L = f'(c)$ if and only if:

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D \Big(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D \Big(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}(f, c, L) \iff \mathrm{FunctionLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\ D,\ c,\ L \right).$$

The predicate $\mathrm{DerivativeAt}(f, c, L)$ asserts that L is the value of the limit of the difference quotient of f at c.

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in D \Big(0 < |x - c| < \delta\ \wedge\ \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \Big).$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{DerivativeAt}(f, c, L) \iff \neg\, \mathrm{FunctionLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\ D,\ c,\ L \right).$$

Remark (Failure modes).

- The difference quotient has no finite limit at c (e.g. $|f(x) - f(c)|/|x - c| \to \infty$).
- The difference quotient oscillates without settling near any single value.
- The proposed value L is the wrong candidate.

Remark (Interpretation). The derivative is a limit of slopes of secant lines. The ε-δ form demands that the difference quotient can be made arbitrarily close to L by taking x sufficiently close to c. The input condition $0 < |x - c|$ is punctured: $x = c$ is excluded because $(f(c) - f(c))/(c - c)$ is $0/0$.

Remark (Dependencies).

- Limit of a Function

Definition (Topological Definition of Derivative at a Point)

Definition 1.1.2 (Topological Definition of Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ and $c \in \operatorname{int}(D)$. f is differentiable at c with derivative L if for every neighbourhood V of L, there exists a neighbourhood U of c such that for all $x \in (U \setminus \{c\}) \cap D$:

$$\frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Standard quantified statement).

$$\forall V \text{ neighbourhood of } L \; \exists U \text{ neighbourhood of } c \; \forall x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \forall V \ni L \; \exists U \ni c \; \forall x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Negated quantified statement).

$$\exists V \text{ neighbourhood of } L \; \forall U \text{ neighbourhood of } c \; \exists x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \notin V.$$

Remark (Negation predicate reading).

$$\neg \, \mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \exists V \ni L \; \forall U \ni c \; \exists x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \notin V.$$

Remark (Failure modes). The topological derivative fails when a fixed output neighbourhood of L cannot be entered by the difference quotient no matter how tightly the input is restricted around c. Geometric reading: no punctured neighbourhood around c maps all secant slopes into any prescribed window around L.

Remark (Interpretation). The topological form replaces ε-balls with abstract neighbourhoods, making the definition coordinate-free and suited for generalisation to metric spaces and manifolds.

Remark (Dependencies).

- Function
- Relative Neighborhood
- Subtraction on $\mathbb{R}$
- Division on $\mathbb{R}$

Definition (Sequential Definition of Derivative at a Point)

Definition 1.1.3 (Sequential Definition of Derivative at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. We say that f is differentiable at c with derivative L if

$$\forall (x_n) \subseteq A \setminus \{c\} \; \big(x_n \to c \Rightarrow \tfrac{f(x_n) - f(c)}{x_n - c} \to L\big).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \;\Rightarrow\; \frac{f(x_n) - f(c)}{x_n - c} \to L.$$

Remark (Definition predicate reading).

$$\text{DerivativeAt}_{\text{seq}}(f, c, L) \iff \forall (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \; \Rightarrow \; \frac{f(x_n) - f(c)}{x_n - c} \to L.$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \; \wedge \; \frac{f(x_n) - f(c)}{x_n - c} \not\to L.$$

Remark (Negation predicate reading).

$$\neg\, \text{DerivativeAt}_{\text{seq}}(f, c, L) \iff \exists (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \; \wedge \; \frac{f(x_n) - f(c)}{x_n - c} \not\to L.$$

Remark (Failure modes). Failure occurs when a single approach sequence $(x_n) \to c$ exists along which the difference quotients either: (i) diverge to infinity (vertical tangent); (ii) oscillate without settling; (iii) converge but to a value other than L (wrong slope candidate). A single such sequence certifies non-differentiability.

Remark (Failure mode decomposition).

$$\underbrace{x_n \to c}_{\text{approach condition}} \quad \wedge \quad \underbrace{\frac{f(x_n) - f(c)}{x_n - c} \not\to L}_{\text{quotient failure}}.$$

The two canonical witness types are: two subsequences with different limiting difference quotients (e.g., $f(x) = |x|$ at $c = 0$), or quotients growing without bound.

Remark (Interpretation). The sequential form is the primary tool for disproving differentiability: exhibit a single sequence $x_n \to c$ along which the difference quotient fails to converge to any single value.

Remark (Dependencies).

- Sequence
- Convergent Sequence
- Derivative at a Point
- Limit of a Function

Theorem (Equivalence of Definitions of the Derivative at a Point)

Theorem 1.1.4 (Equivalence of Definitions of the Derivative at a Point)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$ be a limit point of A. The following are equivalent:*

1. *f is differentiable at c in the ε-δ sense with derivative L.*
2. *f is differentiable at c in the topological sense with derivative L.*
3. *f is differentiable at c in the sequential sense with derivative L.*

Go to proof.

Remark (Standard quantified statement).

$$\text{(i)} \iff \text{(ii)} \iff \text{(iii)}.$$

Remark (Predicate reading).

$$\mathrm{DerivativeAt}(f, c, L) \iff \mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \mathrm{DerivativeAt}_{\mathrm{seq}}(f, c, L).$$

All three predicates name the same underlying condition on the difference quotient of f at c.

Remark (Negated quantified statement).

$$\begin{aligned} \neg\text{(i)} &\iff \neg\text{(ii)} \iff \neg\text{(iii)} \\ &\iff \exists (x_n) \subseteq A \setminus \{c\} : \ x_n \to c \ \wedge \ \frac{f(x_n) - f(c)}{x_n - c} \not\to L. \end{aligned}$$

Remark (Failure modes). The equivalence cannot produce contradictory verdicts: if c is a limit point of A, all three forms agree. Degenerate case: if c is not a limit point of A, the punctured-ball and sequence conditions are vacuously satisfied by every candidate L simultaneously, making the equivalence trivially true for all L rather than characterizing a unique derivative.

Remark (Interpretation). The three forms are notational restatements of the same underlying limit condition on the difference quotient. Analysts choose whichever form is most convenient for the proof at hand.

Remark (Dependencies).

- Derivative at a Point
- Topological Definition
- Sequential Definition

Proposition (Derivative: h-Form Equivalence)

Proposition 1.1.5 (Derivative: h-Form Equivalence). *Let $f : A \to \mathbb{R}$ and $c \in \mathrm{int}(A)$. Then f is differentiable at c with derivative L if and only if*

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = L.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \iff \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = L.$$

Remark (Negated quantified statement).

$$\neg \, \mathrm{DerivativeAt}(f, c, L) \iff \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \neq L.$$

Remark (Failure modes). The h-form fails in two branches: (i) The incremental ratio $(f(c+h)-f(c))/h$ has no finite limit as $h \to 0$ (corner, cusp, or vertical tangent at c). (ii) The ratio converges to a finite value different from L (proposed L is the wrong slope).

Remark (Interpretation). The h-form substitutes $x = c + h$ in the difference quotient. It is the standard notation in calculus and makes the incremental structure of the derivative explicit.

Remark (Dependencies).

- Derivative at a Point

Theorem (Differentiable Implies Continuous)

Theorem 1.1.6 (Differentiable Implies Continuous)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, then f is continuous at c.*

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivableAt}(f, c) \;\Rightarrow\; \text{ContinuousAt}(f, c).$$

Remark (Predicate reading).

$$\text{DifferentiableAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

Remark (Failure modes). The implication can only fail in the reverse direction: continuity at c does not force differentiability at c.

Remark (Failure mode decomposition). The converse fails: $f(x) = |x|$ is continuous at 0 but not differentiable there.

Remark (Contrapositive quantified statement).

$$\neg\text{ContinuousAt}(f, c) \;\Rightarrow\; \neg\text{DifferentiableAt}(f, c).$$

Remark (Contrapositive predicate reading).

$$\neg\,\text{ContinuousAt}(f, c) \implies \neg\,\text{DifferentiableAt}(f, c).$$

Remark (Interpretation). Differentiability is strictly stronger than continuity: the existence of a well-defined tangent slope forces the function values to coalesce at $f(c)$ as $x \to c$.

Remark (Dependencies).

- Derivative at a Point
- Continuity at a Point

Theorem (Uniqueness of the Derivative)

Theorem 1.1.7 (Uniqueness of the Derivative)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, its derivative is unique.*

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \ \wedge\ \text{DerivativeAt}(f, c, L') \ \Rightarrow\ L = L'.$$

Remark (Predicate reading).

$$\text{DerivativeAt}(f, c, L) \ \wedge\ \text{DerivativeAt}(f, c, L') \implies L = L'.$$

Remark (Failure modes). The uniqueness proof assumes $L \neq L'$ and takes $\varepsilon = |L - L'|/2 > 0$. The definition supplies δ making the difference quotient within ε of both L and L' simultaneously, forcing $|L - L'| < 2\varepsilon = |L - L'|$, a contradiction. The cluster-point hypothesis on c is essential: without it, the punctured-ball condition is vacuous and the limit is trivially satisfied by every value.

Remark (Contrapositive quantified statement).

$$L \neq L' \ \Rightarrow\ \neg\big(\text{DerivativeAt}(f, c, L) \ \wedge\ \text{DerivativeAt}(f, c, L')\big).$$

Two distinct candidates cannot both be the derivative of f at c.

Remark (Contrapositive predicate reading).

$$L \neq L' \implies \neg\,\text{DerivativeAt}(f, c, L) \ \vee\ \neg\,\text{DerivativeAt}(f, c, L').$$

Remark (Interpretation). Uniqueness licenses writing $f'(c)$ as a definite value rather than a set of candidates.

Remark (Dependencies).

- Derivative at a Point

Proposition (Derivative of the Identity)

Proposition 1.1.8 (Derivative of the Identity)**.** *Let $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(c) = 1$ for every $c \in \mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall c \in \mathbb{R} : \ \frac{d}{dx}[x]\Big|_{x=c} = 1.$$

Remark (Interpretation). The identity function has slope 1 everywhere. It is the base case for computing derivatives of polynomials via the algebra of derivatives.

Remark (Dependencies).

- Derivative at a Point
- Function

1.2 Secant Tangent

Secant and Tangent — toolkit

Concept family: The geometric objects underlying the derivative: secant lines and tangent lines.

Atomic items in this section:

- Definition: Secant Line
- Definition: Tangent Line

Definition (Secant Line)

Definition 1.2.1 (Secant Line)**.** Let $f : A \to \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Remark (Standard quantified statement)*.*

$$\forall x_1, x_2 \in A \ (x_1 \neq x_2 \Rightarrow \text{SecantLine}(f, x_1, x_2) \text{ has slope } \tfrac{f(x_2)-f(x_1)}{x_2-x_1}).$$

Remark (Interpretation)*.* With increments $\Delta x := x_2 - x_1$ and $\Delta y := f(x_2) - f(x_1)$, the secant slope is $\Delta y/\Delta x$ — the average rate of change of f over the interval $[x_1, x_2]$. This is an exact algebraic identity; no limit is involved. As $x_2 \to x_1$, if the secant slope converges, its limit is the derivative $f'(x_1)$ — the instantaneous rate of change. Geometrically, the secant line rotates about the fixed point $(x_1, f(x_1))$ as x_2 approaches x_1, and the limit position is the tangent line.

Definition (Tangent Line)

Definition 1.2.2 (Tangent Line)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and suppose that f is differentiable at c. The *tangent line* to the graph of f at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

Remark (Standard quantified statement)*.*

$$\text{DifferentiableAt}(f, c) \ \Rightarrow \ \text{LinearApproximationAt}(f, c) = f(c) + f'(c)(x - c).$$

Remark (Interpretation)*.* The tangent line is the limit of the family of secant lines through $(c, f(c))$ and $(x, f(x))$ as $x \to c$. More precisely: the slope $m_{\text{sec}}(c, x) = (f(x) - f(c))/(x - c)$ converges to $f'(c)$, so the tangent line is the line whose slope is that limit. It is also the unique linear function $\ell(x) = f(c) + f'(c)(x - c)$ satisfying $\ell(c) = f(c)$ and $\ell'(c) = f'(c)$ —

the best first-order approximation to f near c. The error $f(x) - \ell(x) = o(x - c)$ as $x \to c$, meaning it vanishes faster than the displacement.

Remark (Dependencies).

- Definition: Secant Line
- Definition: Derivative at a Point

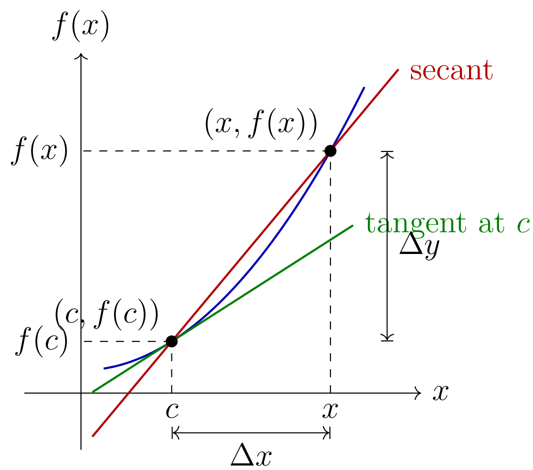

Figure 1.1: The secant line through $(c, f(c))$ and $(x, f(x))$ has slope $\Delta y/\Delta x$. As $x \to c$, the secant line rotates toward the tangent line at $(c, f(c))$, whose slope is $f'(c)$.

1.3 One Sided Derivatives

One-Sided Derivatives — toolkit

Concept family: Left-hand and right-hand derivatives, and their relation to full differentiability.

Atomic items in this section:

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Theorem: Differentiability and One-Sided Derivatives

Definition (Left-Hand Derivative)

Definition 1.3.1 (Left-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a left-hand limit point of I. We say that f has a *left-hand derivative* at c if the limit

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement)*.*

$$\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I \Big(-(\delta) < x - c < 0 \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

In this case $f'_-(c) = L$.

Remark (Definition predicate reading)*.*

$$\mathrm{LeftDerivativeAt}(f, c, L) \iff \mathrm{LeftLimit}\Big(\lambda x.\, \frac{f(x) - f(c)}{x - c},\ c,\ L \Big).$$

Remark (Negated quantified statement)*.*

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I \Big(-\delta < x - c < 0\ \wedge\ \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \Big).$$

Remark (Negation predicate reading)*.*

$$\forall L \in \mathbb{R}\ \neg\, \mathrm{LeftDerivativeAt}(f, c, L).$$

No finite L is the left-hand limit of the difference quotient.

Remark (Interpretation)*.* The left-hand derivative $f'_-(c)$ is the limit of the difference quotient as x approaches c *strictly from the left*. It measures the instantaneous rate of change of f at c as seen from the left side. The right-hand derivative $f'_+(c)$ is the symmetric object from the right. At interior points of an interval, both one-sided derivatives are defined, and the full derivative $f'(c)$ exists if and only if they agree. At endpoints, only the inward one-sided derivative is accessible, and it serves as the boundary derivative. The canonical example: for $f(x) = |x|$ at $c = 0$, $f'_-(0) = -1$ and $f'_+(0) = +1$; since these disagree, $f'(0)$ does not exist.

Remark (Dependencies)*.*

- Definition: Derivative at a Point

Definition (Right-Hand Derivative)

Definition 1.3.2 (Right-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a right-hand limit point of I. We say that f has a *right-hand derivative* at c if the limit

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I \left(0 < x - c < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

In this case $f'_+(c) = L$.

Remark (Definition predicate reading).

$$\mathrm{RightDerivativeAt}(f, c, L) \iff \mathrm{RightLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\ c,\ L\right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I \left(0 < x - c < \delta\ \wedge\ \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \right).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R}\ \neg\, \mathrm{RightDerivativeAt}(f, c, L).$$

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Left-Hand Derivative

Theorem

Theorem 1.3.3 (Differentiability and One-Sided Derivatives)**.** *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \to \mathbb{R}$. Then f is differentiable at $c \in I$ if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f, c) \iff \Big(\mathrm{LeftDerivativeAt}(f, c, L) \wedge \mathrm{RightDerivativeAt}(f, c, L)\Big) \text{ for some } L \in \mathbb{R}.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{DifferentiableAt}(f, c) \iff \neg\Big(\mathrm{LeftDerivativeAt}(f, c, L) \wedge \mathrm{RightDerivativeAt}(f, c, L)\Big) \text{ for all } L \in \mathbb{R}.$$

That is: either a one-sided derivative fails to exist, or both exist but are unequal.

Remark (Interpretation). This theorem decomposes differentiability at an interior point into two one-sided conditions. It is the standard tool for proving non-differentiability: to show f is not differentiable at c, compute $f'_-(c)$ and $f'_+(c)$ and observe they disagree. For $f(x) = |x|$ at $c = 0$: $f'_-(0) = -1 \neq +1 = f'_+(0)$, so f is not differentiable at 0. The theorem also provides the correct definition of differentiability on a closed interval $[a, b]$: differentiable on the open interior (a, b) in the full sense, plus $f'_+(a)$ and $f'_-(b)$ both exist at the endpoints.

Remark (Dependencies).

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Definition: Derivative at a Point

1.4 Algebra Of Derivatives

Algebra of Derivatives — toolkit

Concept family: The pointwise algebra rules for differentiation at a point, and their global (interval) forms.

Pointwise rules (at a point c):

- Theorem: Constant Multiple Rule
- Theorem: Sum Rule
- Theorem: Product Rule
- Theorem: Quotient Rule
- Corollary: Finite Sum Rule
- Corollary: Extended Product Rule
- Corollary: Power Rule for Integer Powers of a Function
- Theorem: Finite Linear Combination Rule

Global rules (on an interval I):

- Theorem: Constant Multiple Rule on an Interval
- Theorem: Sum Rule on an Interval
- Theorem: Product Rule on an Interval
- Theorem: Quotient Rule on an Interval
- Corollary: Finite Sum Rule on an Interval
- Corollary: Extended Product Rule on an Interval
- Corollary: Power Rule for Integer Powers of a Function on an Interval
- Theorem: Finite Linear Combination Rule on an Interval

Theorem (Constant Multiple Rule)

Theorem 1.4.1 (Constant Multiple Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $\alpha \in \mathbb{R}$. If f is differentiable at c, then αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\Rightarrow\; \text{DerivativeAt}(\alpha f,\, c,\, \alpha f'(c)).$$

Remark (Interpretation). Multiplying a function by a constant scales its derivative by the same constant. The proof is one line: the scalar α factors out of the difference quotient algebraically, then the Constant Multiple Rule for Limits passes it through the limit. Differentiation is therefore homogeneous of degree one with respect to scalar multiplication.

Remark (Dependencies).

- Definition: Derivative at a Point
- Constant Multiple Rule for Limits

Theorem (Sum Rule)

Theorem 1.4.2 (Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then $f + g$ is differentiable at c, and*

$$(f + g)'(c) = f'(c) + g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\wedge\; \text{DifferentiableAt}(g, c) \;\Rightarrow\; \text{DerivativeAt}(f + g,\, c,\, f'(c) + g'(c)).$$

Remark (Interpretation). The derivative of a sum is the sum of the derivatives. The proof is immediate: the difference quotient of $f + g$ splits linearly into the sum of the individual difference quotients, and the Sum Rule for Limits distributes the limit across the sum. Together with the Constant Multiple Rule, this establishes that differentiation is a linear operation: $(\alpha f + \beta g)' = \alpha f' + \beta g'$.

Remark (Dependencies).

- Definition: Derivative at a Point
- Sum Rule for Limits

Theorem (Product Rule)

Theorem 1.4.3 (Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then fg is differentiable at c, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f,c) \wedge \mathrm{DifferentiableAt}(g,c) \;\Rightarrow\; \mathrm{DerivativeAt}(fg,\, c,\, f'(c)g(c)+f(c)g'(c)).$$

Remark (Interpretation). The derivative of a product is not simply the product of the derivatives. The correct formula adds two terms: the rate of change of f weighted by the current value of g, plus the rate of change of g weighted by the current value of f. The proof inserts the auxiliary term $f(x)g(c)$ to decouple the increments of f and g, isolating the individual difference quotients. Continuity of f at c (which follows from differentiability) is used to evaluate $\lim_{x\to c} f(x) = f(c)$ in one of the factors.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits
- Product Rule for Limits

Theorem (Quotient Rule)

Theorem 1.4.4 (Quotient Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. Suppose that f and g are differentiable at c and that $g(c) \neq 0$. Then f/g is differentiable at c, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f,c) \wedge \mathrm{DifferentiableAt}(g,c) \wedge g(c) \neq 0 \;\Rightarrow\; \mathrm{DerivativeAt}\left(\frac{f}{g},\, c,\, \frac{f'(c)g(c)-f(c)g'(c)}{(g(c))^2}\right).$$

Remark (Interpretation). The quotient rule is the product rule applied to $f \cdot g^{-1}$, but stated directly. The hypothesis $g(c) \neq 0$ ensures the quotient is defined near c — continuity of g (from its differentiability) and the quotient rule for limits guarantee $g(x) \neq 0$ in a deleted neighbourhood of c, so the difference quotient of f/g is well-defined throughout. The add-and-subtract technique with $f(c)g(c)$ decouples the increments of f and g in the numerator, exactly as in the product rule proof.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits

- Product Rule for Limits
- Quotient Rule for Limits

Corollary

Corollary 1.4.5 (Finite Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\sum_{i=1}^n \alpha_i f_i$ is differentiable at c, and*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(c) = \sum_{i=1}^n \alpha_i f_i'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \in \{1, \ldots, n\} \ \mathrm{DifferentiableAt}(f_i, c) \ \Rightarrow \ \mathrm{DerivativeAt}\left(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c)\right).$$

Remark (Dependencies).

- Theorem: Sum Rule
- Theorem: Constant Multiple Rule

Corollary

Corollary 1.4.6 (Extended Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\prod_{i=1}^n f_i$ is differentiable at c, and*

$$\left(\prod_{i=1}^n f_i\right)'(c) = \sum_{k=1}^n f_k'(c) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(c).$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\forall i \ \mathrm{DifferentiableAt}(f_i, c) \ \Rightarrow \ \mathrm{DerivativeAt}\Big(\textstyle\prod_i f_i,\ c,\ \sum_k f_k'(c) \prod_{i \neq k} f_i(c)\Big).$$

Remark (Dependencies).

- Theorem: Product Rule

Corollary

Corollary 1.4.7 (Power Rule for Integer Powers of a Function)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $n \in \mathbb{N}$. If f is differentiable at c, then f^n is differentiable at c, and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f,c) \ \wedge \ n \in \mathbb{N} \ \Rightarrow \ \text{DerivativeAt}(f^n,\ c,\ n(f(c))^{n-1}f'(c)).$$

Remark (Dependencies).

- Corollary: Extended Product Rule

Theorem

Theorem 1.4.8 (Finite Linear Combination Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \to \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\forall i \ \text{DifferentiableAt}(f_i, c) \ \Rightarrow \ \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c))\,.$$

Remark (Interpretation). The Finite Linear Combination Rule is the full statement that differentiation is a *linear operator*: it commutes with scalar multiplication and addition simultaneously, for any finite collection of differentiable functions. This is not a separate theorem from the Constant Multiple and Sum Rules — it is their inductive closure. Stated in operator language: if $\mathcal{D}(I)$ denotes the vector space of functions differentiable at c, then $D_c : \mathcal{D}(I) \to \mathbb{R}$ defined by $D_c(f) = f'(c)$ is a linear functional. The Finite Linear Combination Rule says $D_c(\sum_i \alpha_i f_i) = \sum_i \alpha_i D_c(f_i)$.

Remark (Dependencies).

- Real-Linear Rule
- Linear Combination of Real-Valued Functions
- Theorem: Constant Multiple Rule
- Theorem: Sum Rule

Theorem

Theorem 1.4.9 (Constant Multiple Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I, then αf is differentiable on I, and*

$$(\alpha f)'(x) = \alpha f'(x) \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \;\operatorname{DifferentiableAt}(f, x) \;\Rightarrow\; \forall x \in I \;\operatorname{DerivativeAt}(\alpha f,\, x,\, \alpha f'(x)).$$

Remark (Dependencies).

- Theorem: Constant Multiple Rule

Theorem

Theorem 1.4.10 (Sum Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then $f+g$ is differentiable on I, and*

$$(f+g)'(x) = f'(x) + g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \;\big(\operatorname{DifferentiableAt}(f, x) \wedge \operatorname{DifferentiableAt}(g, x)\big) \;\Rightarrow\; \forall x \in I \;\operatorname{DerivativeAt}(f+g,\, x,\, f'(x)+g'(x)).$$

Remark (Dependencies).

- Theorem: Sum Rule

Theorem

Theorem 1.4.11 (Product Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then fg is differentiable on I, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \;\big(\operatorname{DifferentiableAt}(f, x) \wedge \operatorname{DifferentiableAt}(g, x)\big) \;\Rightarrow\; \forall x \in I \;\operatorname{DerivativeAt}(fg,\, x,\, f'(x)g(x)+f(x)g'($$

Remark (Dependencies).

- Theorem: Product Rule

Theorem

Theorem 1.4.12 (Quotient Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \big(\text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x) \wedge g(x) \neq 0\big) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}\Big(\tfrac{f}{g},\ x,\ \tfrac{f'(x)g(x)-f}{(g(x))}$$

Remark (Dependencies).

- Theorem: Quotient Rule

Corollary

Corollary 1.4.13 (Finite Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable on I, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \ \forall x \in I \ \text{DifferentiableAt}(f_i, x) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x)).$$

Remark (Dependencies).

- Corollary: Finite Sum Rule

Corollary

Corollary 1.4.14 (Extended Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \ldots, f_n : I \to \mathbb{R}$. If each f_i is differentiable on I, then $\prod_{i=1}^{n} f_i$ is differentiable on I, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(x) = \sum_{k=1}^{n} f_k'(x) \prod_{\substack{i=1 \\ i\neq k}}^{n} f_i(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \ \forall x \in I \ \text{DifferentiableAt}(f_i, x) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}\Big(\textstyle\prod_i f_i,\ x,\ \sum_k f_k'(x) \prod_{i\neq k} f_i(x)\Big).$$

Remark (Dependencies).

- Corollary: Extended Product Rule

Corollary

Corollary 1.4.15 (Power Rule for Integer Powers of a Function on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I, then f^n is differentiable on I, and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \text{DifferentiableAt}(f, x) \ \wedge \ n \in \mathbb{N} \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(f^n,\ x,\ n(f(x))^{n-1} f'(x)).$$

Remark (Dependencies).

- Corollary: Power Rule for Integer Powers of a Function

Theorem

Theorem 1.4.16 (Finite Linear Combination Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i \right)' (x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \ \forall x \in I \ \text{DifferentiableAt}(f_i, x) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x)).$$

Remark (Interpretation). The pointwise statements above — each proved at a single point c — extend to global statements when the hypotheses hold at every point of I. If f is differentiable on I, then $f' : I \to \mathbb{R}$ is itself a function and differentiation acts as a linear operator on the vector space $\mathcal{D}(I)$ of functions differentiable on I:

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

These equalities hold as functions on I. The Finite Linear Combination Rule in its global form is precisely the statement that $D : \mathcal{D}(I) \to \mathcal{F}(I)$ defined by $D(f) = f'$ is a linear map between function spaces, where $\mathcal{F}(I)$ denotes functions on I. The interval variants are not separate theorems — they are the pointwise results collected into function notation by applying the pointwise result at each $x \in I$.

Remark (Dependencies).

- Real-Linear Rule

- Linear Combination of Real-Valued Functions
- Theorem: Finite Linear Combination Rule

Remark (Inverse function — forward note). When f is strictly monotone and differentiable on I with $f'(x) \neq 0$ for all $x \in I$, the inverse $g : J \to I$ is differentiable on $J = f(I)$ and

$$g' = \frac{1}{f' \circ g},$$

as functions on J. This is the global form of the Inverse Function Theorem for derivatives and is treated in the inverse function section.

1.5 Chain Rule

Carathéodory and Chain Rule — toolkit

Concept family: The Carathéodory factorization of the derivative, its characterization theorem, and the Chain Rule.

Atomic items in this section:

- Definition: Carathéodory Factorization
- Theorem: Carathéodory Characterization of Differentiability
- Theorem: Chain Rule

Definition (Carathéodory Factorization)

Definition 1.5.1 (Carathéodory Factorization)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that f admits a *Carathéodory factorization* at c if there exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, f is differentiable at c and $f'(c) = \phi(c)$.

Remark (Standard quantified statement).

$$\exists \phi : I \to \mathbb{R}\ \big(\mathrm{ContinuousAt}(\phi, c)\ \wedge\ \forall x \in I\ [f(x) = f(c) + (x - c)\phi(x)]\big).$$

Remark (Definition predicate reading). The existence of a Carathéodory factorization at c is equivalent to $\mathrm{DifferentiableAt}(f, c)$ by the characterization theorem below. The factorizing function is explicit:

$$\phi(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c} & x \neq c, \\ f'(c) & x = c. \end{cases}$$

Differentiability of f at c is precisely the statement that this piecewise-defined ϕ is continuous at c.

Remark (Interpretation). The Carathéodory factorization replaces the difference quotient — a function with a removable discontinuity at c — with a function ϕ that is genuinely continuous at c. The factorization $f(x) - f(c) = (x-c)\phi(x)$ holds everywhere on I, including at $x = c$ where both sides equal zero. The derivative is recovered as $\phi(c)$ without ever dividing by zero. This reframing converts analytical limit conditions into topological continuity conditions, which compose and factor more cleanly. It is the preferred tool for proving the product rule, quotient rule, and chain rule, because continuity of a product or composition is easier to establish than the existence of a limit of a quotient.

Carathéodory vs. Lipschitz. Both control the increment $f(x) - f(c)$ by the factor $(x-c)$, but differently. The Carathéodory factorization requires the slope function $\phi(x)$ to extend *continuously* to c; it controls the *behaviour* of the slope, not just its size. The Lipschitz condition $|f(x)-f(c)| \le L|x-c|$ only bounds the size of the increment. Carathéodory implies Lipschitz: if ϕ is continuous at c it is bounded near c, so $|f(x)-f(c)| = |\phi(x)||x-c| \le K|x-c|$ for some K near c. The converse fails: $f(x) = |x|$ is Lipschitz on $\mathbb{R}$ with constant 1 but the slope function takes values ± 1 on either side of 0 with no continuous extension.

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Continuous at a Point

Theorem

Theorem 1.5.2 (Carathéodory Characterization of Differentiability). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

(i) f is differentiable at c.

(ii) There exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x-c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$. *Go to proof. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} \text{DifferentiableAt}(f, c) \iff\ & \exists \phi : I \to \mathbb{R}\ \big(\text{ContinuousAt}(\phi, c) \\ & \wedge\ \forall x \in I\ [f(x) = f(c) + (x-c)\phi(x)]\big). \end{aligned}$$

In this case $\phi(c) = f'(c)$.

Remark (Interpretation). This theorem establishes that differentiability and the existence of a Carathéodory factorization are exactly the same condition. The proof is a two-direction construction: in the forward direction, define ϕ as the difference quotient extended by $f'(c)$ at c and verify continuity from the definition of the derivative; in the backward direction, recover the difference quotient from ϕ and read off its limit as $\phi(c)$. The theorem's power is that it converts every future differentiability argument into a continuity argument: to show $g \circ f$ is differentiable, find a continuous Carathéodory function for it; composing the Carathéodory functions of f and g yields one automatically.

Remark (Dependencies).

- Definition: Carathéodory Factorization
- Definition: Derivative at a Point
- Definition: Continuous at a Point

Theorem

Theorem 1.5.3 (Chain Rule). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \to \mathbb{R}$ and $g : J \to \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at c, and*

$$(g \circ f)'(c) = g'(f(c))\, f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\mathrm{DifferentiableAt}(f, c) \ \wedge \ \mathrm{DifferentiableAt}(g, f(c)) \\ &\Rightarrow \ \mathrm{DerivativeAt}(g \circ f,\ c,\ g'(f(c)) \cdot f'(c)). \end{aligned}$$

Remark (Interpretation). The chain rule computes the derivative of a composition by multiplying the derivatives at the respective points: the outer function g is differentiated at the intermediate value $f(c)$, and the inner function f is differentiated at c. The Carathéodory proof is the cleanest: let ϕ_f and ϕ_g be the Carathéodory functions of f at c and g at $f(c)$ respectively. Then

$$g(f(x)) - g(f(c)) = \phi_g(f(x)) \cdot (f(x) - f(c)) = \phi_g(f(x)) \cdot \phi_f(x) \cdot (x - c).$$

The function $x \mapsto \phi_g(f(x)) \cdot \phi_f(x)$ is continuous at c — the product of two continuous functions — and its value at c is $g'(f(c)) \cdot f'(c)$. By the characterization theorem, $g \circ f$ is differentiable with the stated derivative. The naive proof via difference quotients fails because $f(x) = f(c)$ for x near c is possible, causing division by zero; Carathéodory avoids this entirely.

Remark (Dependencies).

- Theorem: Carathéodory Characterization of Differentiability
- Theorem: Differentiable Implies Continuous
- Theorem: Composition of Limits
- Theorem: Product Rule for Function Limits

1.6 Derivative Geometry

Derivative Geometry — toolkit

Concept family: Relative extrema, the Interior Extremum Theorem, and the necessary condition for extrema.

Atomic items in this section:

- Definition: Relative Minimum
- Definition: Relative Maximum
- Definition: Relative Extremum
- Theorem: Interior Extremum Theorem
- Theorem: Necessary Condition for an Interior Extremum
- Corollary: Necessary Condition for an Extremum

Understanding Derivatives — toolkit

Concept family: Monotonicity, local behavior, and qualitative consequences of the derivative.

Atomic items in this section:

- Definition: Increasing at a Point
- Definition: Decreasing at a Point
- Theorem: Zero Derivative Implies Constant
- Corollary: Equal Derivatives Imply Difference Is Constant
- Theorem: Nondecreasing Iff Nonnegative Derivative
- Theorem: Nonincreasing Iff Nonpositive Derivative
- Lemma: Positive Derivative Implies Locally Increasing
- Lemma: Negative Derivative Implies Locally Decreasing
- Theorem: First Derivative Test: Relative Maximum
- Theorem: First Derivative Test: Relative Minimum
- Theorem: Darboux's Theorem

Cross-references (canonical definitions in the functions chapter):

- Definition: Strictly Increasing Function
- Definition: Strictly Decreasing Function

Definition (Relative Minimum)

Definition 1.6.1 (Relative Minimum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative minimum* at c if there exists $\delta > 0$ such that

$$f(c) \leq f(x) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(c) < f(x) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative minimum* at c.

Remark (Standard quantified statement)*.*

$$\exists \delta > 0 \; \forall x \in A \; \big(|x-c| < \delta \implies f(c) \leq f(x)\big).$$

Remark (Definition predicate reading).

$$\text{LocalMinimum}(f, c) \iff \exists \delta > 0\ \forall x \in A\ \big(\text{Within}(x, c, \delta) \implies f(c) \le f(x)\big).$$

Remark (Interpretation). f has a relative minimum at c if $f(c)$ is the smallest value of f in some open neighbourhood of c within the domain. The word *relative* (or *local*) signals that we are comparing $f(c)$ only to nearby values, not to all values of f globally. The strict variant excludes the degenerate case where nearby points share the minimum value. The symmetric notion — relative maximum — reverses the inequality. Together they constitute a relative extremum. These are purely neighbourhood conditions; no derivative appears in the definitions.

Remark (Dependencies).

- Function
- Subset
- Open Interval
- Order on $\mathbb{R}$

Definition (Relative Maximum)

Definition 1.6.2 (Relative Maximum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative maximum* at c if there exists $\delta > 0$ such that

$$f(x) \le f(c) \qquad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(x) < f(c) \qquad \text{for all } x \in \big(A \cap (c - \delta, c + \delta)\big) \setminus \{c\},$$

then f has a *strict relative maximum* at c.

Remark (Standard quantified statement).

$$\exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \implies f(x) \le f(c)\big).$$

Remark (Definition predicate reading).

$$\text{LocalMaximum}(f, c) \iff \exists \delta > 0\ \forall x \in A\ \big(\text{Within}(x, c, \delta) \implies f(x) \le f(c)\big).$$

Remark (Dependencies).

- Definition: Relative Minimum

Definition (Relative Extremum)

Definition 1.6.3 (Relative Extremum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative extremum* at c if f has either a relative minimum at c or a relative maximum at c.

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \iff \text{LocalMinimum}(f, c) \lor \text{LocalMaximum}(f, c).$$

Remark (Definition predicate reading).

$$\text{LocalExtremum}(f, c) \iff \text{LocalMinimum}(f, c) \lor \text{LocalMaximum}(f, c).$$

Remark (Dependencies).

- Definition: Relative Minimum
- Definition: Relative Maximum

Theorem

Theorem 1.6.4 (Interior Extremum Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and f is differentiable at c, then*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \land \text{DifferentiableAt}(f, c) \Rightarrow \text{DerivativeAt}(f, c, 0).$$

Remark (Contrapositive quantified statement).

$$\text{DerivativeAt}(f, c, L) \land L \neq 0 \Rightarrow \neg\,\text{LocalExtremum}(f, c).$$

If $f'(c)$ exists and is nonzero, then c is not a relative extremum.

Remark (Contrapositive predicate reading).

$$\text{DerivativeAt}(f, c, L) \land L \neq 0 \Rightarrow \neg\,\text{LocalMinimum}(f, c) \land \neg\,\text{LocalMaximum}(f, c).$$

Remark (Interpretation). The theorem gives a necessary condition for interior extrema under differentiability: the derivative must vanish. The geometric picture is immediate — at a local max or min, the tangent line must be horizontal, since the function is neither consistently increasing nor consistently decreasing. The proof is a sign argument: at a relative maximum, the difference quotient is non-positive for $x > c$ and non-negative for $x < c$; since the derivative is a single limit, it is squeezed to zero. The converse fails: $f'(c) = 0$ does not imply an extremum ($f(x) = x^3$ at $c = 0$). Critical points — where $f'(c) = 0$ or $f'(c)$ does not exist — are candidates for extrema, not guarantees.

Remark (Dependencies).

- Definition: Relative Extremum
- Definition: Derivative at a Point

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Theorem: Differentiability and One-Sided Derivatives

Theorem

Theorem 1.6.5 (Necessary Condition for an Interior Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \;\Rightarrow\; \neg\,\mathrm{DifferentiableAt}(f, c) \;\vee\; \mathrm{DerivativeAt}(f, c, 0).$$

Remark (Interpretation). This is a weakening of the Interior Extremum Theorem that drops the differentiability hypothesis. Without assuming $f'(c)$ exists, the conclusion is a disjunction: either the derivative fails to exist at c, or it exists and equals zero. This is the correct form for identifying *critical points* — points that are candidates for extrema. The set of critical points is the union of points where f' is zero and points where f' does not exist.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem

Corollary

Corollary 1.6.6 (Necessary Condition for an Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then*

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \;\Rightarrow\; \mathrm{DerivativeAt}(f, c, 0) \;\vee\; \neg\,\mathrm{DifferentiableAt}(f, c).$$

Remark (Dependencies).

- Theorem: Necessary Condition for an Interior Extremum

Definition (Increasing at a Point)

Definition 1.6.7 (Increasing at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *increasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \le f(c) \le f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0 \; \forall x \in A \cap (c-\delta, c) \; \forall y \in A \cap (c, c+\delta) \; \big(f(x) \le f(c) \le f(y)\big).$$

Remark (Definition predicate reading). f is increasing at c if c lies between the values of f on either side in some punctured neighbourhood: f is smaller to the left and larger to the right (weakly). This is a strictly local notion — it says nothing about f outside the δ-neighbourhood of c. A function can be increasing at c without being increasing (globally or on any interval). The non-strict inequalities permit the function to be flat on one or both sides of c, distinguishing this from the notion of a strict local increase.

Remark (Dependencies).

- Definition: Strictly Increasing Function (functions chapter)

Definition (Decreasing at a Point)

Definition 1.6.8 (Decreasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *decreasing at* c if there exists $\delta > 0$ such that for all

$$x \in A \cap (c-\delta, c) \quad \text{and} \quad y \in A \cap (c, c+\delta),$$

we have $f(x) \ge f(c) \ge f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0 \; \forall x \in A \cap (c-\delta, c) \; \forall y \in A \cap (c, c+\delta) \; \big(f(x) \ge f(c) \ge f(y)\big).$$

Remark (Dependencies).

- Definition: Increasing at a Point

Theorem (Zero Derivative Implies Constant)

Theorem 1.6.9 (Zero Derivative Implies Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = 0 \qquad \text{for all } x \in (a, b),$$

then f is constant on I. *Go to proof.*

Remark (Standard quantified statement).

$$\forall x \in (a, b) \; \mathrm{DerivativeAt}(f, x, 0) \;\Rightarrow\; \exists C \in \mathbb{R} \; \forall x \in I \; (f(x) = C).$$

Remark (Contrapositive quantified statement).

$$\neg\big(\exists C \in \mathbb{R} \; \forall x \in I \; (f(x) = C)\big) \;\Rightarrow\; \exists x \in (a, b) \; \neg \mathrm{DerivativeAt}(f, x, 0).$$

If f is not constant on I, then f' is not identically zero on (a, b).

Remark (Contrapositive predicate reading).

$$\neg\,\text{ConstantOn}(f, I) \;\Rightarrow\; \exists x \in (a,b)\ (f'(x) \neq 0).$$

Remark (Interpretation). The MVT is the engine here: if $f'(x) = 0$ everywhere on (a, b), then for any two points $x_1 < x_2$ in I, the MVT supplies a c with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$, so $f(x_1) = f(x_2)$. Since this holds for all pairs, f is constant. The theorem is the foundation of antiderivative uniqueness: two antiderivatives of the same function differ by a constant. The contrapositive is used in ODE uniqueness arguments — if a solution is not identically the zero solution, its derivative is not identically zero somewhere.

Remark (Dependencies).

- Theorem: Mean Value Theorem

Corollary

Corollary 1.6.10 (Equal Derivatives Imply Difference Is Constant)**.** *Let* $I := [a, b] \subseteq \mathbb{R}$*, and let* $f, g : I \to \mathbb{R}$ *be continuous on* I *and differentiable on* (a, b)*. If*

$$f'(x) = g'(x) \qquad \textit{for all } x \in (a,b),$$

then there exists $C \in \mathbb{R}$ *such that* $f(x) = g(x) + C$ *for all* $x \in I$*.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall x \in (a,b)\ (f'(x) = g'(x)) \;\Rightarrow\; \exists C \in \mathbb{R}\ \forall x \in I\ (f(x) = g(x) + C).$$

Remark (Contrapositive quantified statement).

$$\forall C \in \mathbb{R}\ \exists x \in I\ (f(x) \neq g(x) + C) \;\Rightarrow\; \exists x \in (a,b)\ (f'(x) \neq g'(x)).$$

If $f - g$ is not a constant function, then f' and g' disagree somewhere.

Remark (Contrapositive predicate reading).

$$\neg\,\text{ConstantOn}(f - g,\ I) \;\Rightarrow\; \exists x \in (a,b)\ (f'(x) \neq g'(x)).$$

Remark (Interpretation). This is the corollary that makes antiderivatives well-defined up to a constant: any two antiderivatives of the same function on an interval differ by a constant. The proof is one line — apply the previous theorem to $h = f - g$, whose derivative is $(f - g)' = f' - g' = 0$.

Remark (Dependencies).

- Theorem: Zero Derivative Implies Constant

Theorem (Nondecreasing Iff Nonnegative Derivative)

Theorem 1.6.11 (Nondecreasing Iff Nonnegative Derivative)**.** *Let* $I \subseteq \mathbb{R}$ *be an interval, and let* $f : I \to \mathbb{R}$ *be differentiable on* I*. Then* f *is nondecreasing on* I *if and only if*

$$f'(x) \geq 0 \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{NondecreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \geq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) < 0) \ \Rightarrow\ \neg\, \text{NondecreasingOn}(f, I).$$

If the derivative is negative at some point, the function is not nondecreasing.

Remark (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) < 0) \ \Rightarrow\ \neg\, \text{NondecreasingOn}(f, I).$$

Remark (Interpretation). The derivative is the instantaneous rate of change; requiring it to be nonnegative everywhere exactly characterises functions that never decrease. The forward direction (f nondecreasing $\Rightarrow f' \geq 0$) follows from the definition: the difference quotient is nonnegative when f is nondecreasing, so the limit is nonnegative. The reverse direction ($f' \geq 0 \Rightarrow$ nondecreasing) is a consequence of the MVT: for $x < y$, the MVT gives $f(y) - f(x) = f'(c)(y - x) \geq 0$.

Note carefully: $f' \geq 0$ characterises *nondecreasing* (weak), not *strictly increasing* (strict). Strict monotonicity requires $f' > 0$ at all but isolated points. The function $f(x) = x^3$ satisfies $f'(0) = 0$ but is strictly increasing; $f' \geq 0$ with equality only at isolated points is sufficient for strict increase, but $f' > 0$ everywhere is not necessary.

Remark (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Derivative at a Point

Theorem

Theorem 1.6.12 (Nonincreasing Iff Nonpositive Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nonincreasing on I if and only if*

$$f'(x) \leq 0 \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{NonincreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \leq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) > 0) \ \Rightarrow\ \neg\, \text{NonincreasingOn}(f, I).$$

Remark (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) > 0) \ \Rightarrow\ \neg\, \text{NonincreasingOn}(f, I).$$

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative

Lemma

Lemma 1.6.13 (Positive Derivative Implies Locally Increasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \wedge L > 0 \;\Rightarrow\; \exists \delta > 0 \, \forall x \in I \, \big(c - \delta < x < c \Rightarrow f(x) < f(c)\big) \wedge \big(c < x < c + \delta \Rightarrow f(x) >$$

Remark (Contrapositive quantified statement).

$$\forall \delta > 0 \, \exists x \in I \, \big((c - \delta < x < c \wedge f(x) \geq f(c)) \vee (c < x < c + \delta \wedge f(x) \leq f(c))\big) \;\Rightarrow\; f'(c) \leq 0.$$

Remark (Contrapositive predicate reading).

$$\neg \, \mathrm{LocallyIncreasingAt}(f, c) \;\Rightarrow\; f'(c) \leq 0.$$

Remark (Interpretation)*.* If the instantaneous rate of change is strictly positive at c, the function is genuinely climbing through c: it is strictly below $f(c)$ immediately to the left and strictly above immediately to the right. The proof is a direct ε-δ argument: take $\varepsilon = f'(c)/2 > 0$; the definition of the derivative supplies δ such that the difference quotient is within ε of $f'(c)$, forcing it to be positive, which then forces the correct sign on $f(x) - f(c)$. This lemma is the key ingredient in the proof of the First Derivative Test.

Remark (Dependencies).

- Definition: Derivative at a Point

Lemma

Lemma 1.6.14 (Negative Derivative Implies Locally Decreasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f,c,L) \land L < 0 \;\Rightarrow\; \exists\delta > 0\, \forall x \in I\, \big(c{-}\delta < x < c \Rightarrow f(x) > f(c)\big) \land \big(c < x < c{+}\delta \Rightarrow f(x) <$$

Remark (Contrapositive quantified statement).

$$\neg\,\text{LocallyDecreasingAt}(f,c) \;\Rightarrow\; f'(c) \geq 0.$$

Remark (Contrapositive predicate reading).

$$\neg\,\text{LocallyDecreasingAt}(f,c) \;\Rightarrow\; f'(c) \geq 0.$$

Remark (Dependencies).

- Lemma: Positive Derivative Implies Locally Increasing

Theorem (First Derivative Test: Relative Maximum)

Theorem 1.6.15 (First Derivative Test: Relative Maximum)**.** *Let $I := [a,b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a,b)$. Suppose f is differentiable on (a,c) and (c,b). If there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq I$ and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c-\delta, c), \qquad f'(x) \leq 0 \quad \text{for all } x \in (c, c+\delta),$$

then f has a relative maximum at c. Go to proof.

Remark (Standard quantified statement).

$$\exists\delta > 0\, \big[(c-\delta, c+\delta) \subseteq I \;\land\; \forall x \in (c-\delta, c)\ (f'(x) \geq 0) \;\land\; \forall x \in (c, c+\delta)\ (f'(x) \leq 0)\big]$$
$$\Rightarrow\; \text{LocalMaximum}(f,c).$$

Remark (Interpretation). If the function is nondecreasing immediately to the left of c and nonincreasing immediately to the right, then c is a local maximum — the function climbs to c and then falls away. The proof integrates: by the nondecreasing characterisation applied on $(c-\delta, c)$, $f(x) \leq f(c)$ for $x < c$; by the nonincreasing characterisation on $(c, c+\delta)$, $f(x) \leq f(c)$ for $x > c$. The sign condition on f' replaces the Interior Extremum Theorem's converse (which did not hold in general) — here we are asserting a sufficient condition, not proving a necessary one. The test is applicable even when $f'(c) = 0$ or $f'(c)$ does not exist.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative
- Theorem: Nonincreasing Iff Nonpositive Derivative
- Definition: Relative Maximum

Theorem

Theorem 1.6.16 (First Derivative Test: Relative Minimum)**.** *Let $I := [a,b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a,b)$. Suppose f is differentiable on (a,c) and (c,b). If there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq I$ and*

$$f'(x) \leq 0 \quad \text{for all } x \in (c-\delta, c), \qquad f'(x) \geq 0 \quad \text{for all } x \in (c, c+\delta),$$

then f has a relative minimum at c. *Go to proof.*

Remark (Standard quantified statement).

$$\exists \delta > 0 \left[(c-\delta, c+\delta) \subseteq I \ \wedge\ \forall x \in (c-\delta, c)\ (f'(x) \leq 0) \ \wedge\ \forall x \in (c, c+\delta)\ (f'(x) \geq 0) \right] \\ \Rightarrow\ \mathrm{LocalMinimum}(f, c).$$

Remark (Dependencies).

- Theorem: First Derivative Test: Relative Maximum
- Definition: Relative Minimum

Theorem (Darboux's Theorem)

Theorem 1.6.17 (Darboux's Theorem)**.** *Let $I := [a,b] \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$ be differentiable on I. If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a,b)$ such that*

$$f'(c) = k.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall k \ \big(\min(f'(a), f'(b)) < k < \max(f'(a), f'(b)) \ \Rightarrow\ \exists c \in (a,b)\ (f'(c) = k)\big).$$

Remark (Contrapositive quantified statement).

$$\forall c \in (a,b)\ (f'(c) \neq k) \ \Rightarrow\ k \leq \min(f'(a), f'(b)) \ \vee\ k \geq \max(f'(a), f'(b)).$$

If f' skips the value k, then k is not strictly between $f'(a)$ and $f'(b)$.

Remark (Contrapositive predicate reading).

$$\neg\, \mathrm{IVP}(f', [a,b]) \ \Rightarrow\ f \text{ is not differentiable on } [a,b].$$

A function whose derivative fails the intermediate value property cannot be a derivative.

Remark (Interpretation). The Intermediate Value Theorem (IVT) says continuous functions have the intermediate value property (IVP). Darboux's theorem says derivatives also have the IVP — even without being continuous. This is a striking rigidity result: the class of functions that can appear as derivatives is severely constrained. A derivative cannot have a jump discontinuity (it may be discontinuous, but only via oscillation). The proof does not use continuity of f'; it uses differentiability of f to construct an auxiliary function $g(x) = f(x) - kx$ whose derivative changes sign, then applies the Extreme Value Theorem and Interior Extremum Theorem to force an interior extremum where $g'(c) = f'(c) - k = 0$.

Remark (Dependencies).

- Theorem: Extreme Value Theorem
- Theorem: Interior Extremum Theorem
- Definition: Derivative at a Point

1.7 Mean Value Theorem

Mean Value Theorem — toolkit

Concept family: Rolle's Theorem and the Mean Value Theorem — the chapter's load-bearing existence results.

Atomic items in this section:

- Theorem: Rolle's Theorem
- Theorem: Mean Value Theorem

Theorem (Rolle's Theorem)

Theorem 1.7.1 (Rolle's Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and satisfies $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \ \wedge \ \forall x \in (a, b) \ \mathrm{DifferentiableAt}(f, x) \ \wedge \ f(a) = f(b) \\ &\quad \Rightarrow \ \exists c \in (a, b) \ \mathrm{DerivativeAt}(f, c, 0).\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \ \wedge \ \mathrm{DifferentiableOn}(f, (a, b)) \ \wedge \ f(a) = f(b) \\ &\quad \Longrightarrow \ \exists c \in (a, b) \ : \ \mathrm{DerivativeAt}(f, c, 0).\end{aligned}$$

Remark (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \mathrm{sgn}(x - 1/2)$ on $[0, 1]$. Then $f(0) = f(1) = -1$, but f has a jump discontinuity at $x = 1/2$. The derivative does not exist anywhere, so there is no horizontal tangent.
- **Differentiability fails:** Let $f(x) = |x - 1/2|$ on $[0, 1]$. Then $f(0) = f(1) = 1/2$, and f is continuous, but f has a corner at $x = 1/2$ where the derivative is undefined. For all $c \neq 1/2$, we have $f'(c) = \pm 1 \neq 0$.

- **Equal endpoints fail:** Let $f(x) = x$ on $[0, 1]$. Then f is continuous and differentiable with $f'(x) = 1 > 0$ everywhere, but $f(0) = 0 \neq 1 = f(1)$.

Remark (Contrapositive quantified statement).

$$\big(\forall c \in (a, b) \ : \ f'(c) \neq 0\big) \ \Rightarrow \ f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b) \vee f(a) \neq f(b).$$

Remark (Contrapositive predicate reading).

$$\begin{aligned} &\big(\forall c \in (a, b) \ : \ \neg\,\mathrm{DerivativeAt}(f, c, 0)\big) \\ &\quad \Longrightarrow \ \neg\,\mathrm{ContinuousOn}(f, [a, b]) \ \vee \ \neg\,\mathrm{DifferentiableOn}(f, (a, b)) \ \vee \ f(a) \neq f(b). \end{aligned}$$

Remark (Interpretation). If a differentiable function returns to the same value at both endpoints of an interval, the derivative must vanish somewhere in between. Geometrically: if the curve starts and ends at the same height, at some interior point the tangent line is horizontal. The proof combines two ingredients — the Extreme Value Theorem guarantees that a continuous function on a closed bounded interval attains its maximum and minimum, and the Interior Extremum Theorem guarantees that at an interior extremum the derivative is zero. If both extrema occur at the endpoints then f is constant and $f' \equiv 0$ throughout. Rolle's Theorem is the special case of the MVT where the secant slope is zero.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem
- Theorem: Extreme Value Theorem
- Definition: Derivative at a Point

Theorem (Mean Value Theorem)

Theorem 1.7.2 (Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\mathrm{ContinuousOn}(f, [a, b]) \ \wedge \ \forall x \in (a, b) \ \mathrm{DifferentiableAt}(f, x) \\ &\quad \Rightarrow \ \exists c \in (a, b) \ \mathrm{DerivativeAt}\Big(f, \ c, \ \tfrac{f(b) - f(a)}{b - a}\Big) \ . \end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned} &\mathrm{ContinuousOn}(f, [a, b]) \ \wedge \ \mathrm{DifferentiableOn}(f, (a, b)) \\ &\quad \Longrightarrow \ \exists c \in (a, b) \ : \ \mathrm{DerivativeAt}\Big(f, \ c, \ \tfrac{f(b) - f(a)}{b - a}\Big) \ . \end{aligned}$$

Remark (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \begin{cases} 0 & \text{if } x \in [0, 1/2) \\ 1 & \text{if } x \in [1/2, 1] \end{cases}$. Then f has a jump discontinuity at $x = 1/2$, and the secant slope is $\frac{f(1)-f(0)}{1-0} = 1$. But $f'(x) = 0$ wherever the derivative exists, so no interior point achieves the average rate of change.

- **Differentiability fails:** Let $f(x) = |x|$ on $[-1, 1]$. Then f is continuous and the secant slope is $\frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0$. But $f'(x) = \pm 1$ everywhere f is differentiable (i.e., for $x \neq 0$), so no interior point has derivative equal to the secant slope.

The MVT is an existence result. It guarantees some c exists but gives no formula for it.

Remark (Contrapositive quantified statement).

$$\left(\forall c \in (a, b) \ : \ f'(c) \neq \frac{f(b) - f(a)}{b - a}\right) \ \Rightarrow \ f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b).$$

Remark (Contrapositive predicate reading).

$$\begin{aligned} &\left(\forall c \in (a, b) \ : \ \neg\,\mathrm{DerivativeAt}\left(f,\ c,\ \tfrac{f(b)-f(a)}{b-a}\right)\right) \\ &\quad \Longrightarrow \neg\,\mathrm{ContinuousOn}(f, [a, b]) \ \vee \ \neg\,\mathrm{DifferentiableOn}(f, (a, b)). \end{aligned}$$

Remark (Interpretation). The MVT asserts that the instantaneous rate of change equals the average rate of change at some interior point. Geometrically: the tangent line at some point c is parallel to the secant line through $(a, f(a))$ and $(b, f(b))$. The proof reduces to Rolle's Theorem by subtracting the secant: define $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$, which satisfies $g(a) = g(b) = f(a)$, then apply Rolle's. The theorem is an existence result, not a construction: it guarantees some c exists but gives no formula for it. Its immediate corollaries — the constant function corollary, the monotonicity corollary, and the Lipschitz bound corollary — each flow from the MVT by a one-line argument and are the theorem's primary applications.

Remark (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point

1.8 Taylor Expansion

Taylor Expansion Toolkit

Concept family: Taylor polynomials, Taylor remainders, and Taylor's theorem.

Atomic items in this section:

- Definition: Taylor Polynomial at a Point
- Definition: Taylor Remainder
- Definition: Maclaurin Polynomial
- Theorem: Taylor's Theorem with Lagrange Remainder
- Corollary: Taylor Expansion with Peano Remainder

Taylor Expansion

Taylor expansion approximates a differentiable function near a point by a polynomial whose coefficients are forced by the derivatives at that point. The polynomial records the local data, while the remainder records the error left after truncating the expansion.

Definition (Taylor Polynomial at a Point)

Definition 1.8.1 (Taylor Polynomial at a Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \le k \le n$. The **Taylor polynomial of order** n for f at a is

$$T_{n,a}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } I \subseteq \mathbb{R},\ a \in I,\ n \in \mathbb{N},\ f : I \to \mathbb{R}.\\
&\text{If } f^{(k)}(a) \text{ exists for every } 0 \le k \le n,\ \text{then}\\
&\qquad \forall x \in I \left(T_{n,a}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \right).
\end{aligned}$$

Remark (Interpretation). The Taylor polynomial is the unique polynomial of degree at most n whose first n derivatives at a agree with those of f. Its coefficients are not chosen by fitting sample points; they are dictated by derivative data at one point.

Remark (Dependencies).

- Derivative at a Point

- Power Rule for Integer Powers of a Function
- Finite Sum Rule

Definition (Taylor Remainder)

Definition 1.8.2 (Taylor Remainder)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have a Taylor polynomial $T_{n,a}f$. The **Taylor remainder of order** n at a is the function $R_{n,a}f : I \to \mathbb{R}$ defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

Remark (Standard quantified statement)*.*

$$\forall x \in I \ \big(R_{n,a}f(x) = f(x) - T_{n,a}f(x)\big).$$

Remark (Interpretation)*.* The remainder is the exact error made by replacing f with its Taylor polynomial. Taylor's theorem is valuable because it gives usable formulas or estimates for this error.

Remark (Dependencies)*.*

- Taylor Polynomial at a Point

Definition (Maclaurin Polynomial)

Definition 1.8.3 (Maclaurin Polynomial)**.** Let $n \in \mathbb{N}$, and let f be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \le k \le n$. The **Maclaurin polynomial of order** n for f is

$$T_{n,0}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k.$$

Remark (Standard quantified statement)*.*

$$\forall x \in I \ \left(T_{n,0}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k\right).$$

Remark (Interpretation)*.* Maclaurin polynomials are Taylor polynomials centered at the origin. They are not a separate approximation theory; they are the centered-at-zero case of the same construction.

Remark (Dependencies)*.*

- Taylor Polynomial at a Point

Theorem (Taylor's Theorem with Lagrange Remainder)

Theorem 1.8.4 (Taylor's Theorem with Lagrange Remainder)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \to \mathbb{R}$. Suppose that $f, f', \ldots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on (a, b). Then, for every $x \in (a, b)$, there exists c strictly between a and x such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a, b)\ \exists c \in (a, x)$$
$$\left(f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \right).$$

Remark (Interpretation). Taylor's theorem says that the error after truncating at degree n has the same shape as the next Taylor term, but with the next derivative evaluated at some intermediate point rather than at the center. The point c is generally not known explicitly; the theorem is most often used to estimate the size of the remainder.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Mean Value Theorem
- Finite Sum Rule
- Power Rule for Integer Powers of a Function

Corollary (Taylor Expansion with Peano Remainder)

Corollary 1.8.5 (Taylor Expansion with Peano Remainder)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives up to order n in a neighbourhood of a. If $f^{(n)}$ is continuous at a, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o\big((x-a)^n\big) \qquad \text{as } x \to a.$$

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x \to a} \frac{f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k}{(x-a)^n} = 0.$$

Remark (Interpretation). The Peano form is a local asymptotic statement. It says that the Taylor polynomial captures all terms through order n, and the remaining error is smaller than $(x-a)^n$ near the center.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Limit of a Function
- Derivative at a Point

Bibliography